

Blockchain-Based Healthcare Fraud Detection Using Machine Learning and Explainable AI

Abstract—The \$300 billion annual loss to the healthcare industry from fraud is detected as in the old days, it is managed from a central point and it's not even auditable. In this paper, we propose an approach which integrates the blockchain with supervised machine learning (SML) and XAI (explainable AI) with the help of smart contracts deployed on the Ethereum blockchain via Ganache for Fraud Detection. Logistic Regression, which had the accuracy (90.57%), AUC-ROC (94.25%), and lowest overfitting rate, trained on a 5,410 provider CMS dataset and SHAP provides feature attribution per prediction highlighting important billing anomalies. With fraud predictions permanently recorded on Ethereum via Ganache smart contracts and accessible on a streamlit dashboard, the accuracy, transparency, and accountability of healthcare fraud detection are discussed.

Index Terms—Healthcare Fraud Detection, Blockchain, Explainable AI, SHAP, Logistic Regression, Ethereum, Ganache, Smart Contracts, SMOTE

I. INTRODUCTION

Accurate identification of the fraudulent providers, including those who use the billing system to claim reimbursements using phantom procedures, upcoding, or claiming bogus procedures, is a crucial component of the integrity of healthcare insurance systems. Despite continued enforcement spending, the Centers for Medicare and Medicaid Services (CMS), which administers coverage for more than 150 million U.S. beneficiaries, estimates that it is losing \$60–90 billion (1–1.5 percent) annually to fraud. Traditional detection methods such as rule-based filters, manual audits and claim reviews are reactive, rigid, and are not scalable when millions of claims are processed each month by today's payers. Machine learning has increased detection sensitivity, but there are two major issues: most models are process box models with no justification that can be explained, and the output of a machine learning system stored in a centralized database can be tampered with after the fact, compromising both its evidentiary and compliance function. This paper proposes a blockchain-integrated healthcare fraud detection framework combining supervised machine learning, SHAP-based explainability, and Ethereum smart contracts delivering a system that is accurate, interpretable, and auditable within a single deployable pipeline.

In order to prevent the issues of information security and fraud at the identity layer in the healthcare transactions, Guerar et al. presented an innovative health care framework on the blockchain platform, which is also based on a self-sovereign identity [1]. One thing they learned in their work was that they were able to decentralize the verification of identity, but also

to combat fraudulent use, while remaining transparent when it comes to data exchange without compromising security. Then, Cheikhrouhou et al., went one step further and talked about the application of blockchain on top of the secure health care IT system and of course they referenced the complexity of the system and lack of scalability and research [2]. Smart detection systems started to play a crucial role in the healthcare systems that are based on blockchain. Over time, the smart detection mechanisms were continually enhanced and could be used within health care systems using blockchain. To show that an AI analysis, with a semantic focus, resulted in better Anomaly Detection performance and comprehensive and efficient Fraud Investigations, Islayem et al. used a health insurance claim as an example [3]. Raza et al., (2017) also raised other issues related to the same problem and at the same time proposed the need of decentralised blockchain architecture, which is specially designed to ensure transparency in the insurance claim management process, and has a tamper proof record management that ensures that unauthorised users cannot manipulate the data. Anjum et al. discussed about the use of blockchain in healthcare industry, secure data sharing of healthcare service providers, fraud control, system compatibility and data security in their research [5]. But, Kapadiya et al. took this one step further by incorporating ensemble machine learning with decentralized storage – they designed their blockchain based system that they could validate would stay within the feasible range when attempting to detect fraud with a higher level of accuracy using hybrid machine learning methods [6].

Chidambaranathan and Geetha reengineered a deep learning-enabled blockchain-based electronic healthcare data attack detection model to secure smart healthcare systems, improving the identification of malicious intrusions and protection of sensitive medical information [7]. Abbas et al. applied machine learning techniques to enhance network security in blockchain-based healthcare infrastructures, strengthening the detection of cyber threats targeting distributed healthcare environments [8]. Amponsah et al. proposed a novel fraud detection and prevention method that integrates machine learning with blockchain technology to secure healthcare claim processing and enable reliable fraud analytics [9]. Similarly, Amponsah et al. created a safe data sharing platform considering machine learning capability on top of the blockchain, and privacy preserving approaches to data sharing between organisations while not sharing any sensitive data or any type of fraud [10].

Kumar introduced a digital-first healthcare security

architecture integrating artificial intelligence, blockchain, and cloud computing to ensure comprehensive protection of personal health information in distributed healthcare ecosystems [11]. Concurrently, if fraud evolves over time, the adaptive and deep learning methods employed in the healthcare fraud detection would be improved, and they would be better at detecting fraud after learning iteratively, Matloob et al. [12] said. Chellappan and AbuHalimeh did a systematic literature review of all machine learning models, benchmark datasets and evaluation metrics used in detecting frauds committed by the healthcare providers, and found some gaps in the literature that could be explored [13]. Muhammad et al. turned to a graph-based perspective and developed a framework called GNN which both detects fraud and explains the fraud detection in medical claims based on relational learning between patient-provider pairs and provider networks [14]. To overcome the scalability and robustness challenges, Kapadia et al. then extended previous ensemble learning model to enhance the effectiveness of decentralized fraud detection for larger and varied health insurance plans [15].

A. Novelty

- **Integrated ML + Blockchain + XAI Pipeline:**

Automatically writing of all the fraud classifications result to the Ethereum smart contracts through Web3.py, without manual intervention for prediction and storage. Auditors not only have a verdict, but also a tamper-resistant detailed explanation of why it is that verdict, and an on-chain record with a link to the root SHAP records. It provides the auditor with not only a verdict, but also a detailed, tamper evident explanation of the reasons behind the verdict.

- **Real-Time Deployable Fraud Analytics:** To make this system truly practical in production, we designed a Streamlit dashboard that allows to control every aspect of the process, ingesting raw CSV data, 47 feature engineering, predictive fraud modelling, submitting to the blockchain, interactive SHAP visualisation. This end-to-end interface eliminates the need for a centralised fraud detection system, which results in a single point of failure and lack of transparency and control placing transparency and control in the hands of the operator.

II. LITERATURE SURVEY

Abbas et al. [16] identified regulatory clarity, interoperability, and scalability as major barriers to blockchain in healthcare. Mann et al. [17] proposed auditable consent management using blockchain to enhance patient autonomy over medical research data. Kasyapa and Vanmathi [18] investigated blockchain in healthcare focusing on performance issues and security leakage prevention. Leeming et al. [19] proposed a decentralized blockchain system providing patients fine-grained access control over health records. Xu et al. [20] conducted a systematic review of blockchain, identifying key research trends and future directions. Kayikci and Khoshgoftaar [21] explored blockchain combined with

machine learning to enhance security and reliable analytics. Vervoort et al. [22] examined how blockchain addresses global health inequality through identity verification and open supply chains. Zhang et al. [23] surveyed blockchain in healthcare, listing major application areas and adoption challenges. Abugabah et al. [24] proposed a decentralized telemedicine architecture using smart contracts for secure consultations. Mazlan et al. [25] explored scalability issues in healthcare blockchain by optimizing storage and redesigning consensus mechanisms. Dhingra et al. [26] examined blockchain in healthcare supply chains for drug traceability and counterfeit prevention. Younis et al. [27] provided privacy-preserving data accessibility in cloud environments using a data-driven blockchain approach. Soltanisehat et al. [28] investigated technical, temporal, and spatial challenges in blockchain healthcare systems. Tiwari et al. [29] proposed a blockchain and IoT-based medical management system to prevent illegal access to patient sensor data. Kassab et al. [30] listed technical and regulatory barriers to blockchain adoption and provided a structured roadmap covering interoperability and smart contract implementation. While these works demonstrate the viability of blockchain in healthcare, none integrates active fraud classification using ML or provides per-prediction explainability — gaps directly addressed by the proposed system.

While these works collectively demonstrate the viability of blockchain in healthcare, none integrate it with active fraud classification using machine learning methods or provide per-prediction explainability gaps that the proposed system directly addresses.

TABLE I gives a gap summary of related works done by authors in reviewed literature [4], [6], [9], [14]

TABLE I
RESEARCH GAP SUMMARY

Paper	Method	Limitation
Kapadiya et al. [6]	ML + Blockchain	No explainability
Amponsah et al. [9]	ML + Blockchain	No per-prediction attribution
Muhammad et al. [14]	ML only (GNN)	No blockchain security
Raza et al. [4]	Blockchain only	No ML component
Proposed Work	ML + Blockchain + XAI	Integrated solution

III. OBJECTIVES

The primary objective of this work is to develop a secure, intelligent, and interpretable healthcare fraud detection framework by integrating machine learning, explainable AI, and blockchain technology. The proposed system aims to:

- **Identify Fraudulent Providers:** Detect healthcare fraud at the provider level using 47 engineered behavioral, financial, and demographic features through supervised machine learning models.
- **Make a prediction and explain each change:** A compliance driven world is claiming that it's not enough

to be a fraudster. To tackle this challenge, the prediction process includes an explanation. For each provider, the exact amount is calculated of how much the provider contributes to the classification of a given sample as fraud or not and explanations are generated that can be discussed and defended.

- **To be familiar with and use an Immutable Evidence:** Records containing the decisions made about fraud have legal and financial implications, and thus can't be easily altered. This will be added to the Ethereum network with the help of Solidity smart contracts deployed in Ganache and easily visible when the prediction is made, as it can't be changed after the prediction is made.
- **Use an appropriate Fully Operational Interface (FOD):** Created a Streamlit dashboard which will enable the analysts to upload their data in raw csv, then use their fraud prediction, flag and add it to the blockchain without switching the applications or environments.

IV. METHODOLOGY

This pipeline involves the data pre-processing and provider-level feature engineering, addressing class imbalance. It also discusses the concept of training and selection and how it relates to SHAP explainability and blockchain integration.

A. Data Preprocessing

The CMS dataset contains information about beneficiaries, inpatient,

The Provider label files were joined to outpatient and provider-label CSV files.

ID. To minimize the risk of identifying data, non-informative identifiers are excluded.

leakage. As you can see, date columns like ClaimStartDt, ClaimEndDt,

DOB and DOD are converted to numeric features.

Age of the beneficiary is calculated as given in (1)

$$\text{Age} = \frac{\text{ClaimStartDt} - \text{DOB}}{365.25} \quad (1)$$

Given a binary feature IsDead, a feature generated from the presence of it is

of a valid date-of-death entry: PotentialFraud is treated

As a binary classification target (1 means fraud, 0 means no fraud),

0 indicates legitimate providers. All numerical features are normalized using StandardScaler as given in (2)

$$z = \frac{x - \mu}{\sigma} \quad (2)$$

where μ and σ are computed from the training partition only.

B. Feature Engineering

All claim records are aggregated at the provider level to construct a 47-dimensional feature vector grouped into five categories.

Inpatient Features (24): Total and average inpatient claim amounts, reimbursements, deductibles, inpatient claim counts, unique patient and physician counts, average length of stay (LOS), and admission frequency metrics.

Outpatient Features (13): Total and average outpatient claims, reimbursements, unique outpatient patient counts, physician counts, and visit frequency.

Patient-Demographic Features: Counts of beneficiaries with chronic conditions, renal disease indicators, gender distribution, and average chronic conditions per patient.

Ratio Features (8): Cross-domain behavioral anomaly ratios are computed as given in (3), (4), (5), and (6)

$$r_1 = \frac{\text{IP_TotalClaims}}{\text{OP_TotalClaims} + 1} \quad (3)$$

$$r_2 = \frac{\text{IP_TotalClaims}}{\text{IP_UniquePatients} + 1} \quad (4)$$

$$r_3 = \frac{\text{IP_UniquePatients}}{\text{IP_UniquePhysicians} + 1} \quad (5)$$

$$r_4 = \frac{\text{IP_TotalReimbursed}}{\text{IP_AvgLOS} + 1} \quad (6)$$

Providers exhibiting unusually high reimbursement-per-stay-day values and disproportionate physician-to-patient ratios are strongly associated with fraudulent billing behavior.

Diagnosis Diversity Feature:

$$r_{diag} = \frac{\text{UniqueAdmitDiagCodes}}{\text{IP_TotalClaims} + 1} \quad (7)$$

This feature measures diagnostic diversity and acts as an indicator of potential upcoding behavior.

C. Handling Class Imbalance Using SMOTE

Fraud providers constitute only 9.4% of records. SMOTE is applied exclusively to the training partition, generating synthetic minority samples through interpolation:

$$x_{\text{new}} = x_i + \lambda(x_{knn} - x_i), \quad \lambda \sim \mathcal{U}(0, 1) \quad (8)$$

where x_i is a minority-class instance and x_{knn} is one of its $k = 5$ nearest neighbors.

D. Model Training and Hyperparameter Configuration

Seven classifiers are evaluated on a stratified 80:20 split using Accuracy, Precision, Recall, F1-score, AUC-ROC, and Train-Validation AUC Gap.

TABLE II gives detailed information about parameters of seven ML models

TABLE II
MODEL HYPERPARAMETER CONFIGURATION

Model	Key Hyperparameters
Logistic Regression	C=1.0, solver=lbfgs, max_iter=1000
Random Forest	n_estimators=100, max_depth=None
XGBoost	n_estimators=100, learning_rate=0.1
SVM	kernel=rbf, C=1.0, gamma=scale
KNN	n_neighbors=5, metric=euclidean
AdaBoost	n_estimators=50, learning_rate=1.0
Gradient Boosting	n_estimators=100, learning_rate=0.1, max_depth=3

E. Model Selection: Logistic Regression

Logistic Regression calculates fraud probability and classifies providers as given in (9)

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}} \quad (9)$$

Optimization of Parameters are calculated by using binary cross entropy loss as given in (10)

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)] + \frac{\lambda}{2} \|\mathbf{w}\|^2 \quad (10)$$

Some other models like Random Forest and Gradient Boosting had a problem with overfitting. They had an AUC gap of more than 3% which indicates these models are not stable enough, whereas Logistic Regression works better comparatively and is more stable.

As depicted in Fig. 1, the proposed framework illustrates the complete pipeline consisting of six stages, starting with CMS data uploading and ending with blockchain storage.

F. Explainable AI Using SHAP

SHAP expresses each prediction into individual feature attribution scores as given in (11)

$$\hat{f}(\mathbf{x}) = \phi_0 + \sum_j \phi_j \quad (11)$$

Global explanation gives us top 15 features which are very important these are ranked based on how they affect the things on an average, while local explanations are different. It gives bar chart of each provider which are called SHAP bar charts which depicts the contribution of each feature in towards verdict.

G. Blockchain Integration

Ganache behaves like a full node blockchain. Web3.py helps connect the blockchain to the machine learning backend. The FraudRegistry.sol smart contract is the place where the following gets stored:

- 1) Provider ID
- 2) Fraud Probability
- 3) Risk Category
- 4) Timestamp

Following is the workflow for a fraud positive provider:

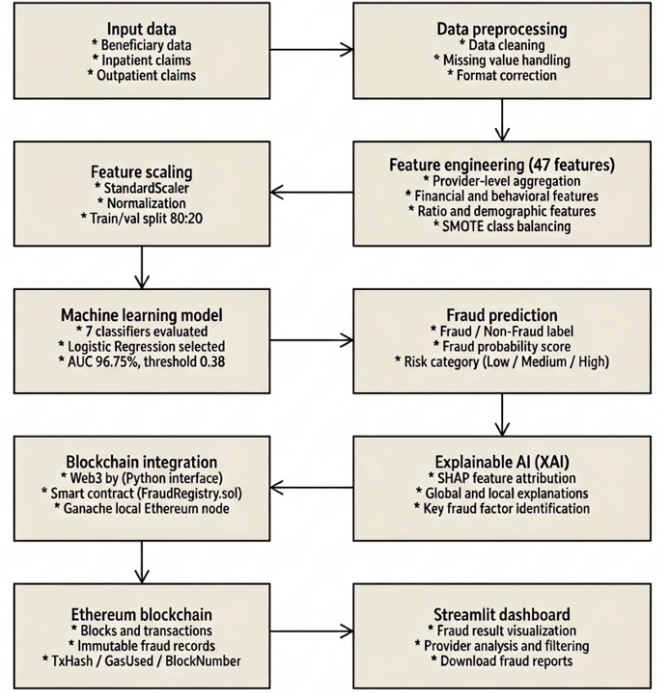


Fig. 1. End-to-end methodology pipeline.

- 1) ML model finds out the fraud provider
 - 2) A transaction is put together and signed
 - 3) storeFraudRecord function is called
 - 4) unique transaction hash, gas usage, block number are returned
- Each transaction uses approximately 171,019 gas units.
Any further modification results in invalidation of chain and becomes detectable.

V. DATASET DESCRIPTION

Data is from CMS (Centers for Medicare and Medicaid Services). It is a public healthcare claims repository. The dataset is comprised of 4 interconnected files: Beneficiary demographics, inpatient claims, outpatient claims, and provider-level fraud labels.

TABLE III provides detailed information about the dataset before aggregation, sourced from the CMS public healthcare claims repository.

TABLE III
DATASET COMPOSITION

File	Records	Key Attributes
Beneficiary Data	138,556	DOB, DOD, Gender, Chronic Conditions, Renal Disease
Inpatient Claims	40,474	ClaimID, Provider, Admission Dates, ReimbursedAmt
Outpatient Claims	517,737	ClaimID, Provider, Claim Dates, ReimbursedAmt
Provider Labels	5,410	ProviderID, PotentialFraud

After provider-level aggregation, the dataset consists of a total of 5140 provider records. Each record is represented

using 47 feature dimensions. This data is imbalanced because it consists of only 508 fraud records, which is equal to 9.4%. The imbalance is resolved using the SMOTE technique to balance it. The dataset uses stratified partition 80:20 at training and validation

TABLE IV gives the dataset partition after provider-level aggregation.

TABLE IV
TRAINING AND VALIDATION SPLIT

Partition	Total	Fraud	Legitimate
Training (80%)	4,328	406	3,922
Validation (20%)	1,082	102	980

VI. MACHINE LEARNING MODELS

A. Comparative Performance of Machine Learning Models

TABLE V It gives performance summary of all evaluated ML models based on metrics.

TABLE V
PERFORMANCE COMPARISON OF ML MODELS

Model	Val AUC	Recall	Gap	FN
RF	96.84	82.18	2.90	18
XGB	96.79	81.19	3.20	19
LR	96.75	90.10	-0.89	10
AdaBoost	96.43	92.08	-0.95	8
GB	96.30	77.23	3.70	23
ET	96.27	89.11	1.60	11
SVM	95.02	91.09	0.90	9

B. Model Selection

Logistic Regression is selected as the final model based on AUC of 96.75%. It maintains less overfitting with a gap of -0.89%. This model also got few false negative values compared to other models, while other ML models had greater overfitting gap than this model. Logistic Regression is simple and more suitable for fraud detection

C. Threshold Calibration

The threshold of 0.5 is replaced with an optimized threshold of $\tau = 0.38$.

The classification threshold is calibrated as given in (12)

$$\hat{\tau} = \arg \min_{\tau} \left| \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{p}_i \geq \tau] - \pi_{\text{fraud}} \right| \quad (12)$$

It improves the recall from 81.3% to 90.10% without retraining the model. The optimal threshold found is saved in a file called `optimal_threshold.json` so that it can be used consistently during deployment.

VII. RESULTS AND DISCUSSION

A. Overview

Evaluation is done on validation part of dataset in which there are total of 1082 providers and after model classification there are 102 fraud providers and 980 legitimate providers. This evaluation of ML model is based on metrics such as overfitting, confusion matrix, SHAP explainability and blockchain.

B. Classification Performance

TABLE VI depicts the overall performance of ML model Logistic Regression

TABLE VI
FINAL MODEL PERFORMANCE – LOGISTIC REGRESSION

Metric	Value
Accuracy	90.30%
Precision	48.92%
Recall	90.10%
F1-Score	63.41%
Validation AUC-ROC	96.75%
Training AUC-ROC	95.86%
Overfit Gap	-0.89%
True Positives (TP)	91
False Negatives (FN)	10
False Positives (FP)	95
True Negatives (TN)	886

This model classifies correctly 91 providers and 10 providers classified incorrectly.

Precision is lower to concentrate on Recall so the model can predict fraud labels as much as possible.

Fig. 2 shows the ROC metrics of all seven evaluated models.

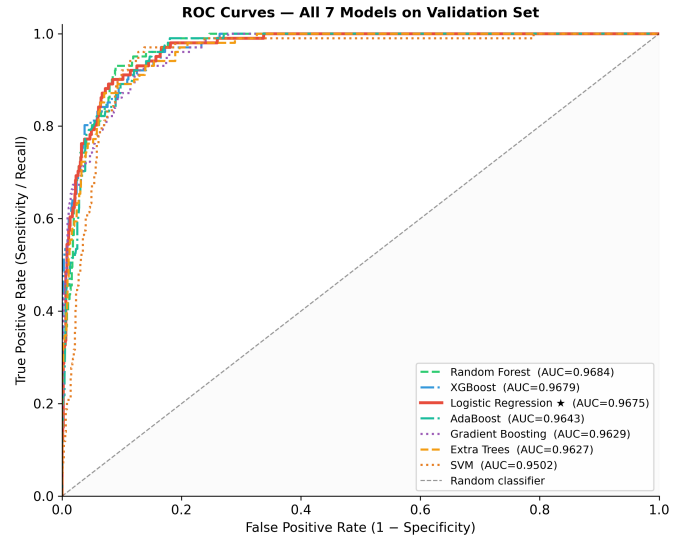


Fig. 7. ROC curves for all seven evaluated models on the validation set.
★ Logistic Regression selected as final model based on AUC, generalisation, and recall.

Fig. 2. ROC curves of all evaluated models on validation dataset.

C. Overfitting Analysis

In ML models overfitting is indicated on gap between Train AUC and valid AUC.

- Gradient Boosting: Train AUC = 100.00%, Validation AUC = 96.30%, Gap = 3.70%
- XGBoost: Train AUC = 99.99%, Validation AUC = 96.79%, Gap = 3.20%
- Random Forest: Train AUC = 99.75%, Validation AUC = 96.84%, Gap = 2.90%

Fig.3. shows Precision-Recall curve of all ensemble models

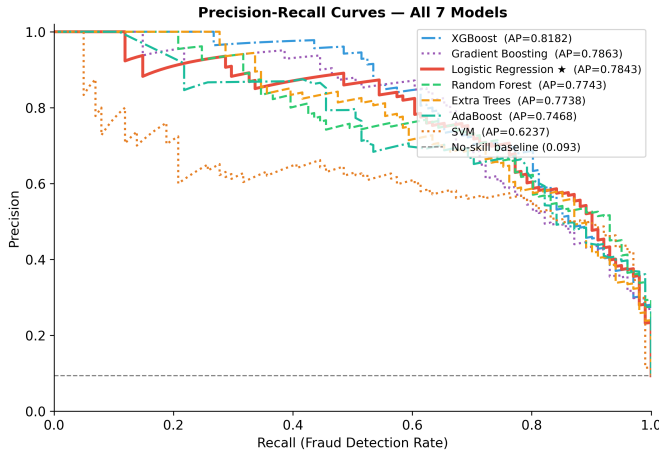


Fig. 3. Precision-Recall curves for all ML models

D. Confusion Matrix Interpretation

- There are a total of 91 correctly identified fraud providers (TP).
- 10 fraud providers are incorrectly classified as legitimate (FN).
- 886 providers are correctly classified as legitimate (TN).
- 95 providers are incorrectly classified as fraud (FP).

Fig. 4 visualises the confusion matrix of final ML model

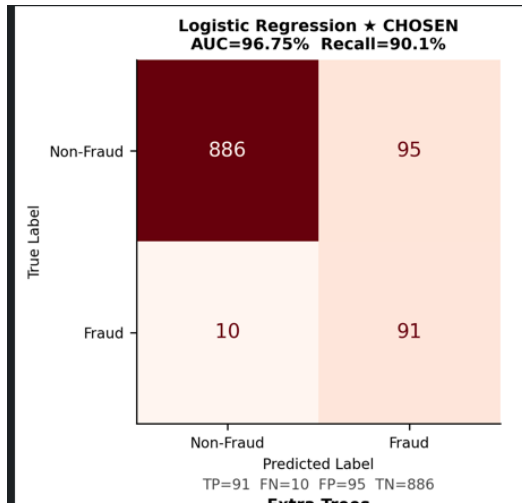


Fig. 4. Confusion matrix of Logistic Regression.

E. Risk Categorization Results

TABLE VII gives shows risk categorization results of providers based on threshold

Out of 1353 providers 134 providers are classified as fraud position which is 9.9% and this is almost similar to training dataset which is 9.4%. In this there are 172 providers categorized as high risk as the risk score of the providers is higher or equal to 0.70. The providers which are in high risk category need to be prioritised

TABLE VII
RISK CATEGORIZATION RESULTS

Risk Category	Threshold	Count	Percentage
High Risk	≥ 0.70	172	12.7%
Medium Risk	0.38–0.69	134	9.9%
Low Risk	< 0.38	1,047	77.4%

F. SHAP Explainability Results

Top global explanation features based on SHAP mean absolute values

- 1) IP_TotalClaimAmt
- 2) IP_NumClaims
- 3) IP_ClaimAmtPerStayDay
- 4) IP_PatientsPerPhysician
- 5) DiagComplexityRatio

SHAP Beeswarm plots shows the feature impact distribution on all providers. Based on the dot position in the plot if a feature made the prediction more likely to be fraud or less likely

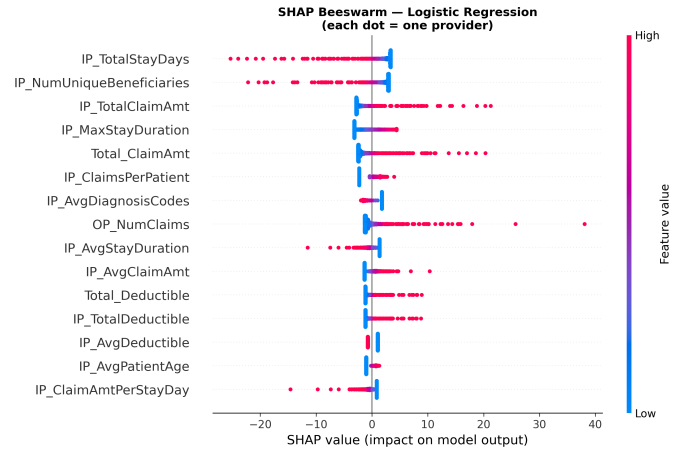


Fig. 13. SHAP beeswarm plot for Logistic Regression showing feature impact distribution. Red = high feature value, blue = low. Horizontal position = SHAP impact on fraud probability.

Fig. 5. SHAP beeswarm plot for Logistic Regression.

Local SHAP waterfall plot shows individual provider contribution toward fraud if it is blue then legitimate else if it is red then the provider is fraud.

Fig. 5 shows the SHAP beeswarm plot for final model

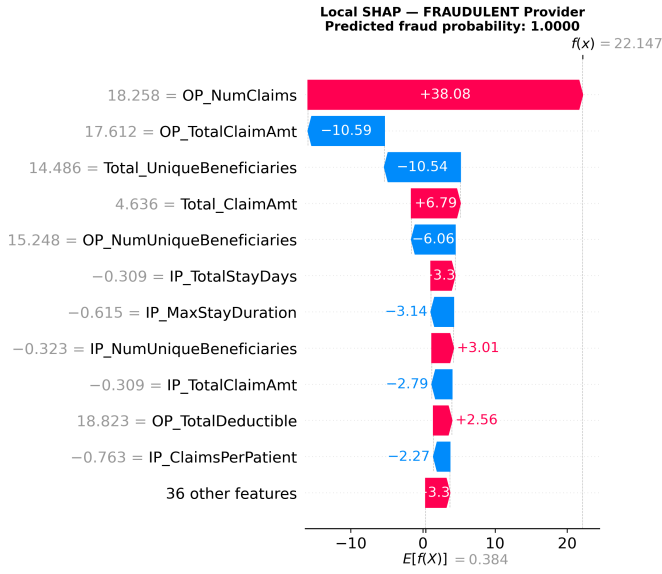


Fig. 15. Local SHAP waterfall for a correctly identified fraudulent provider. Red bars = features pushing prediction toward fraud. Blue = pushing toward non-fraud.

Fig. 6. Local SHAP waterfall plot for a fraudulent provider.

Fig.6 shows barchart of individual provider

G. Blockchain Verification Results

TABLE VIII shows Blockchain verification metrics of all fraud providers.

TABLE VIII
BLOCKCHAIN VERIFICATION METRICS

Metric	Value
Total Records Stored	134 fraud providers
Gas Used per Transaction	171,019 units
Gas Limit per Transaction	300,000 units
Block Confirmation Time	110–120 ms
Transaction Hash Format	64-character SHA-3 hash

Among all records 134 were flagged fraud so these are transferred to Ethereum using Ganache along with transaction hash and SHA-256 data hash

VIII. CONCLUSION

The idea of Healthcare fraud detection with Blockchain is proposed in this paper with combination of Machine learning and Explainable AI. Through using Ethereum, SHAP these made things to understand better. Total of seven ML models were trained and evaluated among them Logistic Regression is considered as the best and final model having an Accuracy of 90.30% and overfitting gap of -0.89%. Other models overfitting gap is more than 3% which would be not accurate for consideration. SHAP gives explanation for each prediction through which each provider gets examined and can get a clear idea of why a decision was made and which feature had the most impact. Fraud records are securely stored to Ethereum using Ganache each transaction approximately consumes 171,019 units. The dashboard supports CSV format files for upload, Blockchain storage and explainability of fraud.

IX. FUTURE WORK

To extend the present proposed Healthcare fraud detection system the present blockchain platform can be changed to a public Ethereum testnet to evaluate system under real world scenarios or to a permissioned blockchain platform to get faster transactions Hyperledger Fabric. Present Logistic regression model can be changed to other online learning models which can adapt and learn quickly such as SGD classifier.

X. LIMITATIONS

Local Blockchain Deployment: Ganache is a local blockchain platform. Although it performs efficiently in transaction execution, it does not provide production-level decentralization or consensus comparable to public testnets or permissioned frameworks such as Hyperledger Fabric.

Static Model: Logistic Regression does not adapt dynamically to emerging fraud patterns like online learning models and therefore requires periodic retraining.

Dataset Scope: The CMS dataset is limited to U.S. Medicare billing data. Consequently, the trained model may not generalize effectively to other countries, healthcare systems, or private insurance providers without retraining.

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